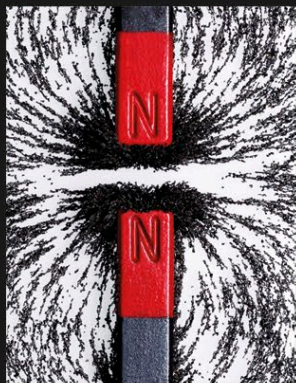


Unlike poles attract, like poles repel

The invisible field of force around a magnet is called a magnetic field. Iron filings show how the magnetic field loops around the magnet from pole to pole.

**Attraction**

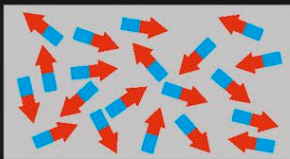
Unlike or opposite poles (a north pole and a south pole) attract each other. Iron filings reveal the lines of force running between unlike poles.

**Repulsion**

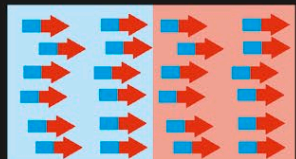
Like poles (two north or two south poles) repel each other. Iron filings show the lines of force being repelled between like poles.

Magnetic induction

An object made of a magnetic material, such as a steel paper clip, is made of regions called domains, each with its own magnetic field. A nearby magnet will align the domain's fields, turning the object into a magnet. The two magnets now attract each other—that is why paper clips stick to magnets. Stroking a paper clip with a magnet can align the domains permanently.

**Domains scattered**

In an unmagnetized object, the domains point in all directions.

**Domains aligned**

When a magnet is nearby, the domain's fields align in the object.

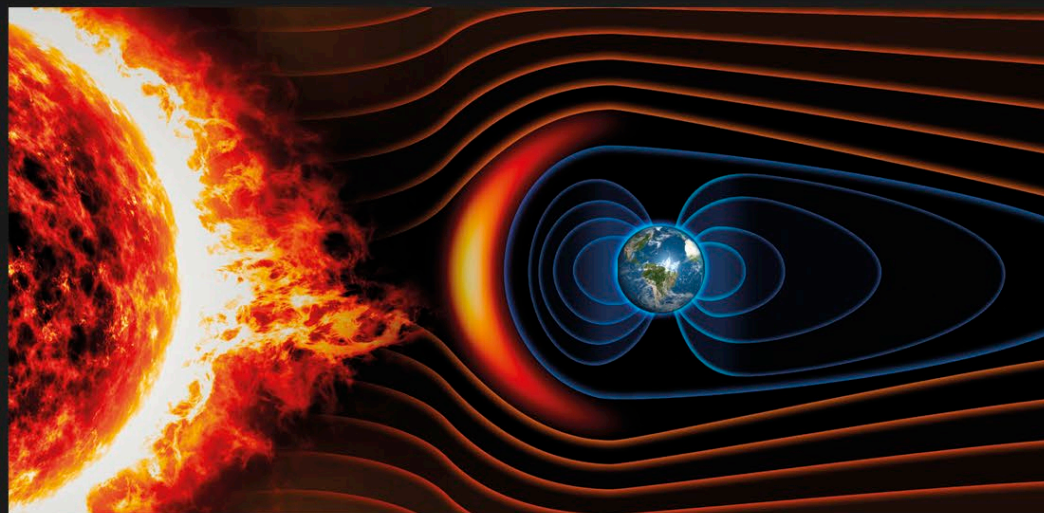
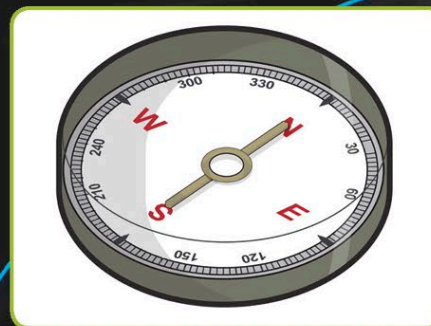
Magnetism

Magnetism is an invisible force exerted by magnets and electric currents. Magnets attract iron and a few other metals, and attract or repel other magnets. Every magnet has two ends, called its north and south poles, where the forces it exerts are strongest.

A magnetic material can be magnetized or will be attracted to a magnet. Iron, cobalt, nickel—and their alloys—and rare earth metals are all magnetic, which means they can be magnetized by stroking with another magnet or by an electric current. Once magnetized, these materials stay magnetic unless demagnetized by a shock, heat, or an electromagnetic field (see p.93). Most other materials, including aluminum, copper, and plastic, are not magnetic.

Magnetic compass

Made of magnetized metal and mounted so that it can spin freely, the needle of a magnetic compass lines up in a north-south direction in Earth's magnetic field. Because the Earth's magnetic North Pole attracts the north, or north-seeking, pole of other magnets, it is in reality the south pole of our planet's magnetic field.

**A teardrop-shaped magnetic field**

Earth's magnetic field protects us from the harmful effects of solar radiation. In turn, a stream of electrically charged particles from the sun, known as the solar wind, distorts the magnetic field into a teardrop shape and causes the auroras—displays of light around the poles (see pp.90–91).

Distortion of the magnetosphere

The stream of charged particles from the sun compresses Earth's magnetic field on the side nearest the sun and draws the field away from Earth into a long "magnetotail" on the far side.